

M2 Contextualization and Tradeoff Analysis: High-Level Synthesis with Game Theoretic Scheduling

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Abstract—[Milestone 2: Scientific Contextualization and Trade-off Analysis. This document maps the game theoretic HLS scheduling and binding approach into the landscape of low-power behavioral synthesis methods. It uses structured comparison tables, tradeoff matrices, and annotated bullet points rather than prose. Benchmark figures used for context are drawn from Murugavel and Ranganathan [1]. AI usage disclosed in the accompanying protocol.]

I. RESEARCH LANDSCAPE: WHERE THIS WORK SITS

A. Three Intersecting Fields

- 1) **HLS / behavioral synthesis**: scheduling, allocation, binding of computational units [5], [4].
- 2) **Low-power design**: minimising switching activity and average power in datapath circuits.
- 3) **Game theory and auction theory**: modelling decentralised, self-interested decision making with the Nash equilibrium as the solution concept [9], [10], [11].

B. Why a Game-Theoretic Formulation

- Scheduling and binding both decide “which unit does which task” — naturally a multi-agent decision problem.
- Functional units behave like self-interested players bidding by power cost; the auction settles on a stable, low-power outcome.
- The Nash equilibrium optimises for all players simultaneously (individual + overall cost), unlike heuristics that optimise one signal.

II. COMPARISON WITH PRIOR LOW-POWER METHODS

A. Method Comparison Table

B. Key Observations

- ILP and LP were the common baselines but were **too complex for large circuits** and optimised scheduling *or* binding *separately*, not both.
- Force-directed scheduling uses a structural concurrency signal, not a direct power-cost decision.
- The game-theoretic method is the only one here that handles **scheduling and binding in one auction framework** while directly minimising power.

- Reported gains over these baselines: **6.3%** (scheduling vs. ILP), **13.9%** (binding vs. LP), **11.8%** (combined vs. ILP) [1].

III. GAME THEORY LANDSCAPE

A. Cooperative vs. Non-Cooperative

- This work uses a **non-cooperative** game: units decide independently, achieving decentralised optimisation.
- Players are fully rational and know the complete game (each other’s preferences and outcomes).

B. Auction Type Selection

- **English / Dutch auctions**: recurring, increasing/decreasing bids — unsuitable, since the power-cost bid is a constant.
- **Second-price sealed-bid**: final value is the second-highest bid — not feasible when the bid value has changed.
- **First-price sealed-bid (chosen)**: sealed, fixed bids; lowest power bid wins — the best fit for the synthesis problem.

IV. TRADEOFF ANALYSIS

A. T1: Power vs. Latency

- Power is the primary objective; latency is *not* co-optimised.
- Cost: a small latency increase — worst case about **2 extra cycles** [1].
- Acceptable for power-critical, throughput-tolerant embedded designs.

B. T2: Power vs. Area

- No additional modules or multiplexers are introduced $\Rightarrow$ **no area overhead**.
- Multiplexer power varies minimally with the binding solution and is small vs. functional-unit power.

TABLE I
LOW-POWER HLS SCHEDULING / BINDING METHODS COMPARED

Method	Type	Scope	Optimises power?	Key limitation
ILP [2]	Exact (integer)	Scheduling	Yes	Too complex for large circuits
LP [2]	Exact (linear)	Binding	Yes	Separate from scheduling
Force-directed [6]	Heuristic	Scheduling	Partially	Structural proxy, no joint binding
Simulated annealing [7]	Stochastic	Sched. + binding	Yes	Slow convergence, randomised
Sehwa (latency) [8]	Heuristic	Scheduling	No	Not power-aware
Game theory (this work)	Auction / Nash	Sched. + binding	Yes	Nash eq. NP-complete for large N

TABLE II
GAME CATEGORIES AND FIT TO HLS BINDING/SCHEDULING

Category	Property	Fit to this work
Cooperative	Players agree on joint strategy; needs central control	Counter to decentralised goal
Non-cooperative	Players act independently; decentralised	Used here

C. T3: Solution Quality vs. Runtime

- Nash equilibrium is NP-complete; complexity grows with players and strategies.
- Mitigated by restricting players to a few units per type (2–3) per game $\Rightarrow$ runtimes **comparable** to ILP/LP.
- Multiple small games per DFG (one per level/control step) instead of one huge game.

D. T4: Scheduling-Only vs. Combined

- Separate scheduling + binding applies the Nash equilibrium *twice*.
- Combined scheduling and binding applies it *once* and gives **better** power reduction.
- Tradeoff: combined formulation is more intricate to set up (single unified payoff matrix).

E. Tradeoff Summary Table

TABLE III
TRADEOFF SUMMARY OF THE GAME-THEORETIC APPROACH

Dimension	Result	Note
Power	$\downarrow$ 6–14%	vs. ILP / LP baselines
Area	No change	No added modules/muxes
Runtime	Comparable	Small player count
Latency	Slight $\uparrow$	≤ 2 extra cycles
Scalability	Limited	NP-complete Nash eq.

V. EMBEDDED SYSTEMS CONSTRAINTS AND CONSEQUENCES

- **Battery life:** directly improved by minimising average power.

- **Thermal / reliability:** lower power $\Rightarrow$ less heating $\Rightarrow$ reliability preserved (50%/10°C rule [3]).
- **Cost:** no extra silicon area; lower cooling cost.
- **Scalability limit:** large circuits make the Nash equilibrium expensive; needs faster computation for big designs.
- **Missing interconnect model:** wire delay / interconnect power not yet captured — relevant in deep-submicron.

VI. JUSTIFICATION OF REFERENCE SCOPE

- **Murugavel & Ranganathan 2003** [1]: the primary game-theoretic behavioral-synthesis reference; source of benchmark comparison data.
- **Shiue & Chakrabarti 2000** [2]: the ILP/LP baseline the approach is measured against.
- **Nash 1951** [9]: foundational solution concept.
- **Owen 1995 / Friedman 1990** [10], [11]: game theory and auction theory background.
- **Gajski 1992 / Benini 2000** [5], [4]: standard HLS and system-level power references.
- **Excluded:** methods optimising only one of scheduling/binding without a power objective are mentioned for contrast but not analysed in depth.

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